Fine-Tuning a Large Language Model (LLM) on Neapolitan Dialect

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Aur Marina Iuliana

Abstract

The project aims to fine-tune a LLaMA model to improve its understanding and generation of text in the Neapolitan dialect. To mitigate training time and GPU memory constraints, Low-Rank Adaptation (LoRA) is used for efficient fine-tuning. Evaluation metrics, including Perplexity, BLEU score, ROUGE score, and BERT score, are employed to compare the Fine-Tuned Model with the Original (Vanilla) version, highlighting challenges and suggesting future work.

1. Introduction and Motivation

In recent years, the latest version of the Large Language Models (LLMs) introduces innovative methodologies that facilitate fine-tuning, significantly enhancing the adaptability of these models across various applications (Santilli & Rodolà, 2023).

While most of LLMs already support the Italian language, evaluating their performance in generating and understanding various Italian dialects could provide valuable insights. Specifically, this project focuses on the **Neapolitan dialect**, as it represents one of the most distinctive Italian dialects.

2. Dataset

The dataset used is the **Identification of Languages and Dialects of Italy (ITDI)** from VarDial 2022 competition. After specifying the path to the XML file containing the dump of the Wikipedia pages, the resulting object consists of a series of `JSON` files that contain the extracted texts.

3. LLaMA Model

The selected model for this project, **`Llama-3.2-1B`**, is a Large Language Model (LLM) designed for multilingual contexts, making it suitable for applications that require understanding and generating text in multiple languages.

Email: Aur Marina Iuliana <aur.1809715@studenti.uniroma1.it>.

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4. Data Preparation and Analysis

4.1. Data Format

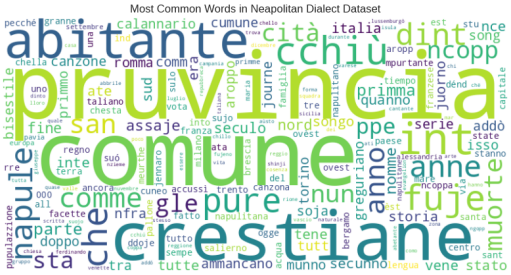
For effective fine-tuning with LLaMA, it is crucial to format the dataset according to a specific recipe (Meta LLaMA). Since the original dataset lacks specific roles and content, the following template is developed to ensure consistency:

```
{
   { "role": "assistant",
     "content": "Describe {title} in Neapolitan dialect." }

   { "role": "user", "content": "{text}" }
}
```

4.2. Exploratory Data Analsys (EDA)

An exploratory data analysis (EDA) is conducted to gain insights into the data. Firstly, we investigate the most common words (Figure 1) in the dataset and their distribution. We found that the dataset is **widely distributed**, as no single word frequency particularly dominates the others.


Most Common Words in Neapolitan Dialect Dataset
abitante
cchiù
dint
ncopp
pruvincia
comune
int
anne
fuje
muorte
crestiane
napule
che
sta
comme
gle
pure
anno
nun
primma
quanno
ppe
serie
assaje
secuolo
fine
san
cumune
calannario
città
italia
canzone
fomma
comm
primmo
bisestile
sud
journe
aropp
tre
juorno
capitale
addò
state
isso
stanno
vene
stato
centro
munno
secunno
ammancano
tutte
tutt
tene
soja
storia
parte
doppo
nfra
napulitana
cuneo
accussì
trento
canzona
torino
ovest
paese
alessandria
regnò
regnò
popolazione
lingua
sant
santa
bergamo
ncoppa
ncoppa
ncoppa

language. We found an average divergence of **0.798**, which suggests that the Neapolitan dialect may not be easily interpreted by the model.

5. Fine-Tuning

5.1. Low-Rank Adaptation (LoRA)

Large Language Models (LLMs) are often high-dimensional and over-parametrized, while the information encoded in them tends to be well-approximated in a much lower dimension (Frankle & Carbin, 2019).

`Low-Rank Adaptation (LoRA)` Sis a Parameter-Efficient Fine-Tuning (PEFT) technique that decreases the number of parameters for fine-tuning by decomposing large matrices. This reduction in parameters results in a decrease in training time and GPU memory usage, while preserving the quality of the outputs.

After applying LoRA with a **rank of 4** and a **scaling factor of 16**, only 0.23% of the parameters are fine-tuned during the training process.

5.2. Fine-Tuning Arguments

The **`SFTTrainer`** is employed for supervised fine-tuning of the LLaMA model. The model is fine-tuned using a **learning rate** of 1×10^{-4}, **weight decay** of 0.01, and AdamW **optimizer**, as they represent a common choice for similar tasks.

5.3. Fine-Tuning Results

The fine-tuning process is configured to run for a maximum of **100 steps** and completes in about **21 minutes**, resulting in a training loss of 1.43 and an **evaluation loss of 2.06**. These values suggest that the model is reasonably fitting the training data but is struggling to generalize effectively on unseen data.

6. Performance Evaluation

6.1. Perplexity

`Perplexity` is one of the most common metrics for evaluating language models. It measures a model's prediction ability by calculating the exponentiated average negative log-likelihood of a sequence, reflecting how well it predicts each token given prior tokens.

6.2. BLEU, ROUGE and BERT Scores

Additionally, BLEU, ROUGE and BERT scores are employed to assess the model's performance:

- **`BLEU score`** measures precision by comparing n-grams in generated and reference text.
- **`ROUGE score`** focuses on recall by assessing the overlap of content.
- **`BERT score`** uses embeddings to evaluate semantic similarity.

These scores are computed using partial test set due to memory and time constraints. To ensure consistency, both the reference response and the generated response are limited to a maximum of 150 characters.

6.3. Fine-Tuned Model versus Vanilla Model

A summary of the models' performance using both the Fine-Tuned Models and Vanilla can be found in Table 1.

We can assert that the Fine-Tuned model achieved better performance, especially regarding **perplexity**, which is significantly lower compared to the Vanilla model one.

The other scores highlight that the model struggles to perfectly reconstruct the reference response. This can be attributed to the fact that the dataset is widely distributed and shows a high divergence from the Italian language.

Table 1. Performance Comparison of the Fine-Tuned and Vanilla LLaMA Models

	bleu	rouge1	rouge2	rougeL	bert_f1	plp
FT	0.0002	0.0872	0.0160	0.0756	0.5668	17.3501
Vanilla	0.0000	0.0544	0.0104	0.0511	0.5473	36.5685

7. Qualitative Assessment and Future Work

Examples of sentences generated by the Fine-Tuned Model can be found in the Notebook. Qualitatively, the sentences produced are in the Neapolitan dialect; however, they do not always maintain consistency.

For this work, a model with only 1 billion parameters is used due to time and memory constraints. Employing a model with 3 billion parameters would probably lead to improved results.

Furthermore, it would be beneficial to experiment with other dialects to investigate if there is a correlation between the average divergence of the dialect from the Italian language and the model's performance.

References

Frankle, J. and Carbin, M. The lottery ticket hypothesis: Finding sparse, trainable neural networks, 2019. URL https://arxiv.org/abs/1803.03635.

Meta LLaMA. Llama recipes. https://github.com/meta-llama/llama-recipes/tree/main.

Santilli, A. and Rodolà, E. Camoscio: an italian instruction-tuned llama, 2023. URL https://arxiv.org/abs/2307.16456.